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Fast Antimicrobial Susceptibility Testing for Gram-Negative Bacteremia
The FAST Randomized Clinical Trial

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IMPORTANCE Novel blood culture diagnostics provide rapid antimicrobial susceptibility testing (AST) results for bacteria causing bloodstream infections (BSIs) but have unclear clinical impact.

OBJECTIVE To evaluate clinical outcomes of blood culture testing using a rapid AST method compared with standard AST in patients with BSIs caused by gram-negative bacilli in regions with high prevalence of antimicrobial resistance.

DESIGN, SETTING, AND PARTICIPANTS Open-label randomized clinical trial that enrolled participants from December 2023 to May 2025 (final follow-up, June 18, 2025) at 7 medical centers in Greece (n = 2), India (n = 1), Israel (n = 3), and Spain (n = 1). Hospitalized patients (adults and children) with BSIs caused by gram-negative bacilli were eligible. Statistical analysis was conducted from August 2025 to January 2026.

INTERVENTION Patients were randomized to undergo blood culture evaluation using rapid, phenotypic AST directly from positive blood cultures plus standard susceptibility testing (n = 413) vs standard susceptibility testing alone (n = 437). Local antimicrobial stewardship teams reviewed all patients and provided treatment recommendations.

MAIN OUTCOMES AND MEASURES The primary outcome was a desirability of outcome ranking (DOOR) at day 30, with 3 categories (alive without deleterious events, alive with deleterious events, and death). The difference between rapid and standard testing was summarized as the probability that the DOOR outcomes were more desirable in the rapid testing group. Superiority was concluded if the lower limit of the 95% CI exceeded 50%. Secondary outcomes included 30-day mortality, length of hospitalization, intensive care unit admission, acquisition of hospital-acquired infections, time to effective antibiotic therapy within 3 days, and time to antibiotic escalation or deescalation within 3 days.

RESULTS Of the 899 patients randomized, 850 were included in the primary outcome analysis (median [IQR] age, 72 [21] years; 43% female). The probability that DOOR outcomes were more favorable in the rapid testing group was 48.8% (95% CI, 45.3%-52.4%). Median time to effective antibiotic therapy was not different between the groups in the as-randomized population. Median time to antibiotic escalation or deescalation was faster in the rapid testing group by 14 hours (95% CI, 6-22) in the as-randomized population. There were no differences between the groups in other secondary outcomes. In the prespecified subgroup with carbapenem-resistant infections, median time to effective therapy was 9.5 hours in the rapid testing group vs 28 hours in the standard testing group (difference, −18 hours [95% CI, −42 to 6]).

CONCLUSIONS AND RELEVANCE Among patients with gram-negative bacilli BSIs, rapid blood culture AST was not superior to standard testing by DOOR. When considered with other efficacy and safety outcomes, these findings may help inform the use of rapid susceptibility testing in clinical practice.

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Gram-negative bacilli (GNB) cause up to half of bloodstream infections (BSIs) worldwide.[1-3] Ineffective therapy for GNB BSIs is associated with mortality rates in excess of 30%, particularly in low- and middle-income settings.[4-10] More rapid determination of the antimicrobial susceptibility characteristics of bacteria causing BSI may lead to earlier pathogen-directed antimicrobial therapy, more optimal antibiotic utilization, less emergence of antimicrobial resistance, and improved clinical outcomes. Although novel and rapid blood culture diagnostics can provide faster organism identification and antimicrobial susceptibility testing (AST) results than standard methods, these diagnostics are expensive and their clinical effect has not been rigorously assessed. Some observational studies suggest a mortality benefit or shorter hospital stay with use of rapid blood culture diagnostics,[11-15] but these results have potential confounding and cannot establish causality, and their findings have not been replicated in limited-resource settings or in prospective clinical trials.

To the study investigators' knowledge, 5 prospective clinical trials have evaluated the clinical effect of rapid AST for BSIs caused by GNB. All 5 trials demonstrated that rapid susceptibility testing resulted in faster organism identification and antimicrobial resistance detection and decreased time to optimal therapy, but did not reduce mortality or shorten duration of hospitalization.[16-21] Limitations of these prior trials are that they were conducted in high-income countries with low prevalence of antimicrobial resistance; most patients were receiving effective antibiotic therapy at enrollment, potentially limiting benefits of the intervention; they were not sufficiently powered to detect differences in clinical outcomes; and they utilized costly diagnostic testing, which is less applicable and generalizable to limited-resource settings.

This study sought to assess the clinical effect of rapid blood culture AST in settings with high endemicity of antimicrobial-resistant GNB, where such rapid diagnostics are most likely to have clinical impact. The primary objective of this pragmatic trial was not to evaluate the rapid testing method for diagnostic accuracy, but to determine whether its implementation in real-world clinical settings with high rates of antimicrobial resistance improved clinical outcomes. A rapid AST method approved for clinical use by regulatory agencies in the countries of participating sites was used.

Key Points

Question Does use of rapid antimicrobial susceptibility testing directly from blood culture bottles positive for gram-negative bacteria improve clinical outcomes compared with standard care?

Findings In this randomized clinical trial of 850 patients with gram-negative bloodstream infections performed in 4 countries with high levels of antimicrobial resistance, the probability of a more desirable outcome at day 30 in the rapid blood culture antimicrobial susceptibility testing group compared with the standard testing group was 48.8%, which did not achieve the prespecified criterion for superiority.

Meaning Rapid blood culture antimicrobial susceptibility testing for gram-negative bacteremia was not superior to standard testing in achieving a more desirable outcome at day 30.

Methods

Design and Oversight

The Fast Antimicrobial Susceptibility Testing for Gram-Negative Bacteremia Trial (FAST) was an open-label, randomized, superiority trial conducted at 7 medical centers in 4 countries with a high prevalence of multidrug-resistant gram-negative bacteria: Greece, India, Israel, and Spain. FAST enrolled patients from December 2023 to May 2025. The institutional review boards from each participating center approved the study. A waiver of informed consent was approved in all but 1 site, where participants provided written informed consent. The waiver was approved at most sites due to the study's minimal-risk nature and because it was not practicable to obtain consent when the diagnostic test needed to be performed as soon as a positive blood culture result was obtained. The study protocol was previously published.[22] An independent data and safety monitoring board supervised the trial. Consolidated Standards of Reporting Trials (CONSORT) reporting guidelines were followed.[23] This trial was conducted in accordance with the ethical principles set out in the Declaration of Helsinki.[24]

Participants

Patients of any age were eligible if they had GNB in blood cultures and were hospitalized at the time of the blood culture result. Patients were excluded for any of the following criteria: (1) blood culture with GNB within the prior 7 days; (2) deceased at time of blood culture result; (3) presence of gram-positive bacilli, gram-positive cocci, gram-negative cocci, yeast, fungi, or multiple morphologies of GNB on Gram stain of blood culture; or (4) prior enrollment in the study. Full inclusion and exclusion criteria are listed in Supplement 1.[22] Participant race and ethnicity were self-reported according to multiple prefixed categories. This information was collected to describe the study population because outcomes can differ across racial and ethnic groups.

Randomization

Participants were randomized 1:1 to either a rapid or standard AST group, stratified by site and intensive care unit status at the time of Gram stain result, using a permuted block size of 4. Screening and randomization were performed by laboratory technologists as soon as possible after a blood culture with GNB on Gram staining was noted in the laboratory. Participants were required to be randomized within 16 hours of blood culture positivity to conform with the rapid test instructions for use.[25] The treating clinicians were unaware of randomization assignment at the time of randomization, so initial antibiotic choice was not affected by group assignment. The stewardship team, laboratory technologists, and outcomes assessors were not masked to group assignment.

Intervention

For participants in both groups, blood culture organism identification was determined using matrix-assisted laser desorption ionization time-of-flight (MALDI-ToF) mass spectrometry. For participants randomized to the rapid testing group, AST was determined by directly testing positive blood culture bottle contents using the VITEK REVEAL (bioMérieux), in addition to standard AST, which was performed on subcultured bacteria using methods that varied by site (eMethods in Supplement 3). Rapid AST results were reported in the participant's medical record and/or to the treating clinician by telephone or secure message, upon availability.

All participants in both groups underwent routine review by institutional antimicrobial stewardship programs that contacted the treating clinician if modifications to therapy were indicated at any time following notification of the Gram stain result, organism identification, and/or AST results. Antibiotic selection, dose, and route were determined by the clinical team, which chose whether to modify treatment based on the study interventions. In both groups, randomization or testing not performed per protocol was considered a protocol deviation.

Primary and Secondary Outcomes

The primary outcome was the desirability of outcome ranking (DOOR) at day 30 after randomization. DOOR is an ordinal composite outcome that integrates both efficacy and safety, comprising 3 ordered categories: alive without deleterious events, alive with at least 1 deleterious event, and death. Deleterious events were assessed by study staff who were not masked to group assignment. The DOOR end point has been previously used in studies of bacteremia.[26-33] Deleterious events were defined as (1) unsuccessful discharge (not discharged from index hospitalization or readmission within 30 days); (2) lack of clinical response (relapse of BSI with the same species causing the index infection, local suppurative complications not present at enrollment, or seeding distant sites with the same infecting organism after enrollment); and (3) undesirable events, including kidney failure or acquisition of hospital-acquired colonization or infection with multidrug-resistant organisms.

Secondary outcomes included 30-day mortality, length of hospitalization up to 30 days, intensive care unit admission after Gram stain result, acquisition of hospital-acquired infections due to a multidrug-resistant organism or *Clostridioides difficile*, time to effective antibiotic therapy within 3 days, and time to antibiotic escalation or deescalation within 3 days. Patients were followed up until death, discharge, or for 30 days after randomization, whichever occurred first. Prespecified exploratory outcomes included a DOOR outcome with 5 categories (alive without deleterious events; alive with 1, 2, or 3 deleterious events; and death), agreement between rapid and standard AST results, and clinician adherence with antimicrobial stewardship recommendations. Outcomes were evaluated in the as-randomized population and prespecified subgroups, including the population with on-panel organisms only (ie, organisms included on the rapid AST). Further details are provided in the eMethods in Supplement 3.

Statistical Analysis and Power Calculation

Statistical analysis was conducted from August 2025 to January 2026. Primary and secondary outcomes were assessed as randomized. This analysis included all randomized patients, except those who were unevaluable for reasons that may not have been known before randomization, including (1) standard culture did not yield GNB (ie, erroneous Gram stain), (2) rapid AST was initiated more than 16 hours after blood culture positivity, (3) deceased prior to randomization, or (4) blood culture with GNB within the preceding 7 days. The study was powered to evaluate for superiority of rapid over standard AST based on the DOOR outcome. Based on recent randomized clinical trials,[18,34] a probability of 55% was estimated, representing the likelihood that DOOR outcomes in the rapid AST group would be more favorable than those in the standard AST group. For this probability to be detected, a sample size of 405 participants per group was calculated to provide at least 80% power to reject the null hypothesis (that the DOOR outcome distributions are identical [ie, the probability equals 50%]) at a 2.5% significance level using a 1-sided Wilcoxon-Mann-Whitney test. Allowing for a 10% unevaluable rate, a total of 900 participants was planned for enrollment. One interim analysis was conducted when 50% of patients were enrolled and had at least 30 days of follow-up. Further details are described in the statistical analysis plan in Supplement 2.

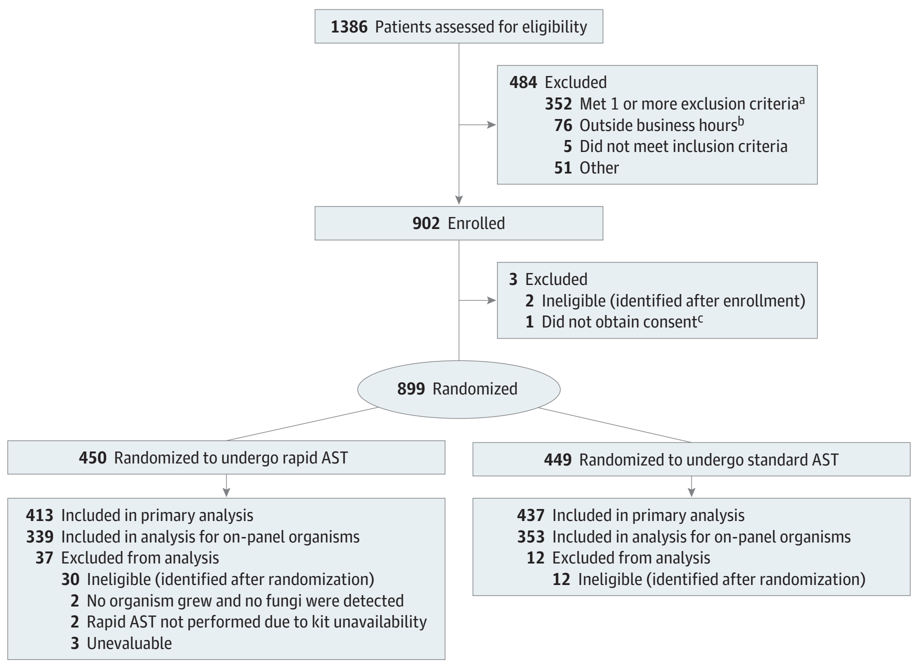
Participants missing day 30 deleterious events were considered not to have experienced the event. The probability that DOOR outcomes in the rapid AST group were more favorable than in the standard AST group was estimated using a Wilcoxon-Mann-Whitney statistic for ordinal outcomes, with the corresponding 95% CI calculated using the Halperin method.[35] Superiority of rapid over standard AST was concluded if the lower limit of the 95% CI exceeded 50%. As a sensitivity analysis, a tiebreaker-based approach was conducted for the DOOR outcome. When participants from both groups fell into the same category, either alive without deleterious events or alive with at least 1 deleterious event, hospital length of stay was used as a tiebreaker, with shorter stays considered more favorable. If both participants fell into the death category, time to death was used as a tiebreaker, with longer survival considered more favorable. For the secondary outcomes of time to effective therapy and time to antibiotic escalation or deescalation, cumulative incidence function plots with the difference in Kaplan-Meier median time to event were used to compare the survival distributions between the 2 groups. The change in the distribution of undertreatment, appropriate treatment, and overtreatment within 3 days after Gram stain was also examined among participants with on-panel monomicrobial infections.

Results

Participant Characteristics

Between December 2023 and May 2025, a total of 899 patients were randomized from 7 sites in Greece, India, Israel,

Figure 1. Flowchart of Patient Recruitment and Randomization in the Fast Antimicrobial Susceptibility Testing (AST) for Gram-Negative Bacteremia Trial (FAST)



GNB indicates gram-negative bacilli.

[a] Blood culture with GNB in prior 7 days, deceased at time of Gram stain result, mixed morphologies on Gram stain, or prior enrollment in this study.

[b] Varied by site.

[c] Consent required by 1 site.

and Spain, with 850 included in the analysis (437 randomized to undergo standard testing alone and 413 to undergo rapid AST) (**Figure 1**). Baseline characteristics were generally balanced between the groups, but patients who were immunocompromised were numerically greater in the rapid testing group and patients with greater severity of illness at enrollment (assessed by hypotension or vasopressor requirement or quick Pitt bacteremia score of ≥2) were numerically greater in the standard testing group (**Table 1**).[36,37] The microbiology of bloodstream isolates was similar between the groups, with the most common organisms being *Escherichia coli* (343 of 917 isolates [37.4%]) and *Klebsiella pneumoniae* (202 of 917 isolates [22.0%]) (eTable 1 in Supplement 3). Cephalosporin or carbapenem resistance was detected in 362 of 820 (44%) patients overall; was numerically greater in the rapid vs standard testing group; and varied by country, being greater at sites in India and Greece than in Israel and Spain (eTable 1 in Supplement 3). At enrollment, nearly two-thirds of patients in both groups were receiving effective antibiotic therapy (Table 1).

Primary Outcome

The probability that DOOR outcomes in the rapid AST group were more favorable than in the standard AST group was 48.8% (95% CI, 45.3%-52.4%), which did not demonstrate superiority of rapid AST over standard AST (**Figure 2**) (eFigure 1 in Supplement 3). The DOOR probability was similar using the tiebreaker approach. The number and proportion of each DOOR event are summarized in eTable 2 in Supplement 3.

Secondary Outcomes

All-cause 30-day mortality, length of hospital stay, and acquisition of multidrug-resistant organisms were similar between groups. The proportion of patients who remained hospitalized at 30 days post enrollment was 38 of 413 (9.2%) in the rapid AST group vs 58 of 437 (13.3%) in the standard AST group in the as-randomized population (difference, −4.1 [95% CI, −8.3 to 0.2]), and in the prespecified subgroup of patients with carbapenem-resistant infections, the corresponding proportions were 13 of 86 (15.1%) vs 21 of 75 (28.0%), respectively (difference, −12.9 [95% CI, −25.6 to −0.2]) (**Table 2**). In the prespecified subgroup of patients with cephalosporin-resistant infections, the median days of hospitalization after enrollment was 8 in the rapid AST group vs 11 in the standard AST group (difference, −3 [95% CI, −6 to 0]) (Table 2). Time to effective antibiotic therapy within 3 days did not differ between groups in the as-randomized population (Table 2, **Figure 3**A). Median hours to first antibiotic escalation or deescalation within 3 days of randomization was 22 in the rapid AST group vs 36 in the standard AST group (difference, −14 [95% CI, −22 to −6]) (Table 2).

Prespecified Exploratory Outcomes

There were no differences between the groups in a 5-category DOOR probability. The proportion of patients with stewardship recommendations within 3 days was greater in the rapid vs standard AST group (difference, 17.7 [95% CI, 11.5-23.9]) (Table 2). Acceptance of stewardship recommendations by treating clinicians was greater in the rapid vs standard AST

Table 1. Baseline Patient Characteristics in the As-Randomized Population

Characteristic	Antimicrobial susceptibility testing, No. (%)	
	Rapid (n = 413)	Standard (n = 437)
Age, median (IQR), y	73 (61-81)	72 (59-81)
Race[a]		
Asian	47 (11)	48 (11)
Black	1 (0)	5 (1)
White	365 (88)	384 (88)
Ethnicity, No./total No. (%)		
Arab	23/411 (6)	27/435 (6)
Hispanic or Latino	30/411 (7)	31/435 (7)
Jewish	115/411 (28)	120/435 (27)
Other[b]	243/411 (59)	257/435 (59)
Sex		
Male	238 (58)	248 (57)
Female	175 (42)	189 (43)
Country of enrollment		
Greece	196 (47)	210 (48)
Israel	143 (35)	148 (34)
India	43 (10)	48 (11)
Spain	31 (8)	31 (7)
Comorbidities		
Age-adjusted Charlson Comorbidity Index score, median (IQR)[c]	5 (3-6)	4 (3-6)
Immunocompromised	106 (26)	98 (22)
Severity of illness		
Quick Pitt bacteremia score ≥2[d]	91 (22)	110 (25)
Intensive care unit at enrollment	69 (17)	76 (17)
Hypotension or vasopressor requirement at enrollment	101 (24)	123 (28)
Microbiology		
On-panel organism[e]	339 (82)	353 (81)
Cephalosporin- or carbapenem-resistant organism[f]	184 (45)	178 (41)
Source of bacteremia		
Urinary tract	139 (34)	160 (37)
Unidentified	126 (31)	129 (30)
Intra-abdominal	77 (19)	71 (16)
Lung	25 (6)	32 (7)
Central venous catheter[g]	26 (6)	28 (6)
Other[h]	14 (3)	18 (4)
Skin or soft tissue	9 (2)	12 (3)
Hospital-acquired infection[i]	215 (52)	236 (54)
Effective antibiotic therapy at baseline[j]	254 (62)	276 (63)

[a] Self-reported race collected from list of prespecified categories.

[b] Self-reported ethnicities were non-Hispanic, non-Arab, and non-Jewish. Collected from fill in the blank field.

[c] Scores range from 0 to 29, with higher scores indicating greater predicted mortality.

[d] Five-point score with 1 point given for each of the following at enrollment: hypothermia (temperature <36 °C), hypotension (systolic blood pressure <90 mm Hg or vasopressor use), respiratory failure (respiratory rate >25 breaths/minute or need for mechanical ventilation), cardiac arrest, or altered mental status. A score of 2 or greater was associated with increased risk of mortality.[36,37]

[e] Included organisms detected on rapid antimicrobial susceptibility testing panel: *Acinetobacter baumannii* (species and complex), *Citrobacter freundii*, *Citrobacter koseri*, *Enterobacter cloacae* (species and complex), *Escherichia coli*, *Klebsiella aerogenes*, *Klebsiella oxytoca*, *Klebsiella pneumoniae* (species and complex), *Proteus mirabilis*, and *Pseudomonas aeruginosa*.

[f] Ceftriaxone, ceftazidime, or cefuroxime; excludes patients whose isolates were not tested against carbapenem or cephalosporin antibiotics.

[g] Included preexisting catheters and those placed during hospitalization.

[h] Infections of bones or joints, cardiovascular or central nervous systems, or surgical site infections.

[i] Positive blood culture collected more than 2 days after admission.

[j] Defined as receipt of at least 1 antibiotic to which the bloodstream isolate was susceptible by standard antibacterial susceptibility testing methods and to which the organism was not intrinsically resistant. Baseline period includes time from Gram stain result through 2 hours after Gram stain result.

group (difference, 14.8 [95% CI, 8.4-21.2]). Categorical agreement between rapid and standard AST methods among patients randomized to the rapid AST group ranged from 58% to 100% for organism-antibiotic combinations (eTable 3 in Supplement 3).

Prespecified Subgroup Analyses

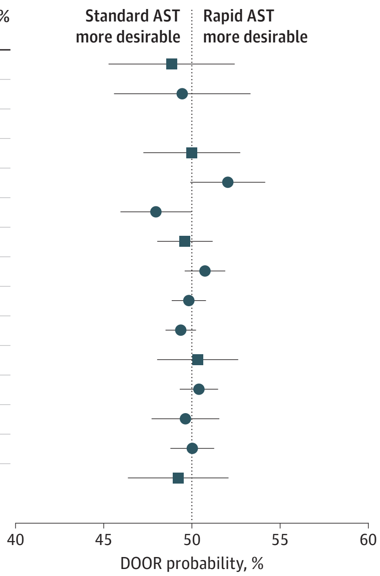
Subgroup analyses demonstrated no evidence of heterogeneity in treatment effect on the DOOR outcome across prespecified subgroups (eFigure 2 in Supplement 3). There were differences by country in antibiotics administered (eTable 4 in Supplement 3). There were differences across sites in standard testing methods, turnaround times, and antimicrobial stewardship personnel and workflows (eTable 5 in Supplement 3). There were also differences across sites in mortality and proportion of patients receiving effective antibiotics within 1 and 3 days. Sites enrolling more patients with antibiotic-resistant infections had the lowest proportions of patients receiving effective antibiotics and the highest mortality (eTable 6 in Supplement 3).

There were differences in antibiotic utilization among prespecified subgroups with antibiotic-resistant organisms. Time to first antibiotic escalation or deescalation within 3 days of randomization was 12 hours in the rapid AST group vs 35 hours in the standard AST group for patients with cephalosporin-resistant infections (difference, −23 [95% CI, −34 to −12]); for carbapenem-resistant infections, the corresponding times were 13 vs 47 hours, respectively (difference, −29 [95% CI, −44 to −14]) (Table 2) (eFigure 3B and C in Supplement 3). Time to effective antibiotic therapy in patients with cephalosporin-resistant infections was 3 vs 4 hours for the rapid vs standard AST groups, respectively (difference, −1 [95% CI, −6 to 4]); for those with carbapenem-resistant infections, the corresponding times were 9.5 vs 28 hours, respectively (difference, −18 [95% CI, −42 to 6]) (Table 2, Figure 3B and C).

The effectiveness and spectrum of antibiotics received were evaluated among 692 participants with monomicrobial

Figure 2. Forest Plot of Primary Outcome in the As-Randomized Population

Outcome	No. (%) Rapid AST (n = 413)	No. (%) Standard AST (n = 437)	DOOR probability, % (95% CI)
DOOR			48.8 (45.3-52.4)
With hospitalization/mortality days tiebreaker			49.5 (45.6-53.3)
Respective component			
Unsuccessful discharge	87 (21.1)	92 (21.1)	50.0 (47.2-52.7)
Not discharged	38 (9.2)	58 (13.3)	52.0 (49.9-54.2)
Readmission	49 (11.9)	34 (7.8)	48.0 (45.9-50.0)
Lack of clinical response	25 (6.1)	23 (5.3)	49.6 (48.0-51.2)
Relapse of bacteremia	9 (2.2)	16 (3.7)	50.7 (49.6-51.9)
Suppurative complications	9 (2.2)	8 (1.8)	49.8 (48.9-50.8)
Seeding distant sites	9 (2.2)	4 (0.9)	49.4 (48.5-50.2)
Undesirable events	54 (13.1)	60 (13.7)	50.3 (48.0-52.6)
Clostridioides difficile infection	9 (2.2)	13 (3.0)	50.4 (49.3-51.5)
New MDRO infection	38 (9.2)	37 (8.5)	49.6 (47.7-51.6)
Kidney failure	14 (3.4)	15 (3.4)	50.0 (48.8-51.3)
Death	100 (24.2)	99 (22.7)	49.2 (46.4-52.1)



DOOR and DOOR with mortality or hospital days as a tiebreaker represent the DOOR probability that a participant in the rapid AST group would experience a more favorable outcome compared with a participant in the standard AST group. In cases where 2 participants achieved the same DOOR mortality (later death was better), hospitalization days (fewer was better) were used as a tiebreaker to provide a more nuanced distinction between them. The probability was estimated using the Wilcoxon-Mann-Whitney statistic. If the 95% CI for the probability excluded the point of 50%, the difference in the DOOR end point between groups was considered statistically significant. The individual components of DOOR—unsuccessful discharge, lack of clinical response, undesirable events, and death—were analyzed in the same manner as DOOR. The median (IQR) length of stay from enrollment to discharge was 8 (11) days for the rapid AST group and 8 (13) days for the standard AST group.

AST indicates antimicrobial susceptibility testing; DOOR, desirability of outcome ranking; MDRO, multidrug-resistant organism.

infections (defined in Supplement 3). The proportions of patients who received undertreatment, appropriate treatment, and overtreatment at baseline were similar between groups. Patients in the rapid AST group moved from the undertreatment or overtreatment categories into the appropriate treatment category at earlier time points compared with patients in the standard AST group (**Figure 4**). Among patients receiving undertreatment at baseline (defined in Supplement 3), the time to antibiotic escalation was 24 hours in the rapid vs 53 hours in the standard AST group (difference, −27 [95% CI not estimable]) (eFigure 4A in Supplement 3). Among patients receiving unnecessarily broad-spectrum antibiotic therapy or overtreatment at baseline, time to antibiotic deescalation was faster in the rapid than the standard AST group and fewer than half of patients underwent deescalation in the standard AST group (eFigure 4B in Supplement 3).

Discussion

In this multinational trial enrolling patients with bacteremia caused by GNB from regions of the world with high antimicrobial resistance, rapid AST was not superior to standard testing based on the DOOR primary outcome. Additionally, there were no significant differences for most secondary outcomes.

However, rapid AST was associated with improvements in more timely antibiotic modifications and targeted antibiotic therapy in certain subsets of patients, more frequent guidance from antimicrobial stewardship teams, and greater acceptance of stewardship recommendations. In the subgroup of patients with carbapenem-resistant infections, rapid AST was also associated with fewer patients remaining hospitalized at day 30; to the study investigators' knowledge, this finding has not been observed in previous diagnostic trials. More successful discharge may stem from faster targeted treatment with use of rapid AST but warrants further study.

Rapid AST did not affect mortality, which occurred in 20% of patients in the primary analysis and 40% of patients with carbapenem-resistant infections, consistent with rates reported in the literature.[38-41] Lack of effect of rapid AST on mortality has been reported in other studies and may occur for many reasons.[16-20] All-cause mortality was assessed, which has complex drivers and may be due to illness severity or comorbidities other than BSI. Moreover, sites enrolling patients with resistant organisms did not always have the most effective antibiotic therapies available. This treatment gap, which has been described in low- and middle-income countries,[42] may contribute to rapid AST not affecting clinical outcomes; it is likely that rapid AST may have had greater benefit if first-line, effective antibiotics had been available at all sites. Importantly, after the trial began, study sites in Greece were unable to

Table 2. Secondary and Exploratory Outcomes in the As-Randomized Population and Prespecified Subgroups

Outcome	As-randomized population			Cephalosporin-resistant organisms[a]			Carbapenem-resistant organisms		
	AST		Difference in proportions or medians (95% CI)	AST		Difference in proportions or medians (95% CI)	AST		Difference in proportions or medians (95% CI)
	Rapid (n = 413)	Standard (n = 437)		Rapid (n = 164)	Standard (n = 154)		Rapid (n = 86)	Standard (n = 75)	
Clinical outcomes (within 30 d following enrollment), No. (%)									
Days from enrollment to discharge, median (95% CI)[b]	8 (7 to 8)	8 (7 to 9)	0 (−1 to 1)	8 (7 to 12)	11 (9 to 13)	−3 (−6 to 0)	11 (6 to 14)	13 (8 to 18)	−2 (−9 to 5)
Hospitalized at 30 d	38 (9.2)	58 (13.3)	−4.1 (−8.3 to 0.2)	20 (12.2)	26 (16.9)	−4.7 (−12.4 to 3.1)	13 (15.1)	21 (28.0)	−12.9 (−25.6 to −0.2)
Intensive care unit admission post enrollment	14 (3.4)	16 (3.7)	−0.3 (−2.8 to 2.2)	7 (4.3)	6 (3.9)	0.4 (−4.0 to 4.7)	5 (5.8)	2 (2.7)	3.1 (−3.0 to 9.3)
Readmission	51 (12.3)	41 (9.4)	3.0 (−1.2 to 7.2)	13 (7.9)	18 (11.7)	−3.8 (−10.3 to 2.8)	3 (3.5)	4 (5.3)	−1.8 (−8.2 to 4.6)
Acquisition of MDRO[c]	55 (13.3)	51 (11.7)	1.6 (−2.8 to 6.1)	28 (17.1)	21 (13.6)	3.4 (−4.5 to 11.3)	20 (23.3)	16 (21.3)	1.9 (−10.9 to 14.8)
Kidney failure	21 (5.1)	34 (7.8)	−2.7 (−6.0 to 0.6)	7 (4.3)	13 (8.4)	−4.2 (−9.5 to 1.2)	1 (1.2)	4 (5.3)	−4.2 (−9.7 to 1.4)
Lack of clinical response[d]	31 (7.5)	34 (7.8)	−0.3 (−3.8 to 3.3)	15 (9.1)	11 (7.1)	2.0 (−4.0 to 8.0)	6 (7.0)	4 (5.3)	1.6 (−5.8 to 9.0)
Relapse of infection	12 (2.9)	24 (5.5)	−2.6 (−5.3 to 0.1)	6 (3.7)	5 (3.2)	0.4 (−3.6 to 4.4)	4 (4.7)	4 (5.3)	−0.7 (−7.4 to 6.1)
Mortality	100 (24.2)	99 (22.7)	1.6 (−4.1 to 7.3)	52 (31.7)	43 (27.9)	3.8 (−6.3 to 13.8)	41 (47.7)	30 (40.0)	7.7 (−7.6 to 23.0)
Antibiotic treatment and stewardship outcomes, No. (%)									
Effective antibiotic therapy at baseline[e]	254 (61.5)	276 (63.2)	−1.7 (−8.2 to 4.9)	80 (48.8)	73 (47.4)	1.4 (−9.6 to 12.4)	33 (38.4)	29 (38.7)	−0.3 (−15.4 to 14.8)
Effective antibiotic therapy within 24 h[e]	345 (83.5)	329 (75.3)	8.2 (2.9 to 13.6)	134 (81.7)	97 (63.0)	18.7 (9.1 to 28.4)	62 (72.1)	35 (46.7)	25.4 (10.7 to 40.2)
Effective antibiotic therapy within 3 d[e]	365 (88.4)	374 (85.6)	2.8 (−1.7 to 7.3)	145 (88.4)	130 (84.4)	4.0 (−3.5 to 11.5)	69 (80.2)	51 (68.0)	12.2 (−1.3 to 25.7)
Hours from Gram stain to effective antibiotic therapy within 3 d, median (95% CI)[e]	0 (NE to NE)	0 (NE to NE)	0	3 (0 to 7)	4 (1 to 7)	−1 (−6 to 4)	9.5 (3 to 13)	28 (1 to 50)	−18 (−42 to 6)
Initial antibiotic change is escalation[f]	193 (46.7)	183 (41.9)	4.9 (−1.8 to 11.5)	94 (57.3)	79 (51.3)	6.0 (−4.9 to 17.0)	52 (60.5)	35 (46.7)	13.8 (−1.5 to 29.1)
Initial antibiotic change is deescalation[f]	116 (28.1)	107 (24.5)	3.6 (−2.3 to 9.5)	35 (21.3)	23 (14.9)	6.4 (−2.0 to 14.8)	16 (18.6)	15 (20.0)	−1.4 (−13.6 to 10.8)
Hours from Gram stain to antibiotic escalation/deescalation, median (95% CI)[f]	22 (14 to 25)	36 (30 to 45)	−14 (−22 to −6)	12 (10 to 18)	35 (27 to 47)	−23 (−34 to −12)	13 (10 to 21)	47 (32 to 57)	−29 (−44 to −14)
Stewardship recommendation made within 3 d	314 (76.0)	255 (58.4)	17.7 (11.5 to 23.9)	130 (79.3)	93 (60.4)	18.9 (9.0 to 28.8)	68 (79.1)	45 (60.0)	19.1 (5.0 to 33.1)
Stewardship recommendation acceptance within 2 d	287 (69.5)	239 (54.7)	14.8 (8.4 to 21.2)	120 (73.2)	88 (57.1)	16.0 (5.7 to 26.4)	61 (70.9)	43 (57.3)	13.6 (−1.1 to 28.3)

Abbreviations: AST, antimicrobial susceptibility testing; MDRO, multidrug-resistant organism; NE, not estimable.

[a] Ceftriaxone, ceftazidime, or cefuroxime.

[b] Kaplan-Meier median and the median time difference were estimated using quantile regression methods for censored data.

[c] Included methicillin-resistant *Staphylococcus aureus*, vancomycin-resistant *Enterococcus* species, third-generation cephalosporin nonsusceptible *Enterobacterales*, carbapenem-resistant *Enterobacterales* or *Acinetobacter* species, *Pseudomonas aeruginosa* resistant to carbapenems or multidrug resistant, or *Candida auris*.

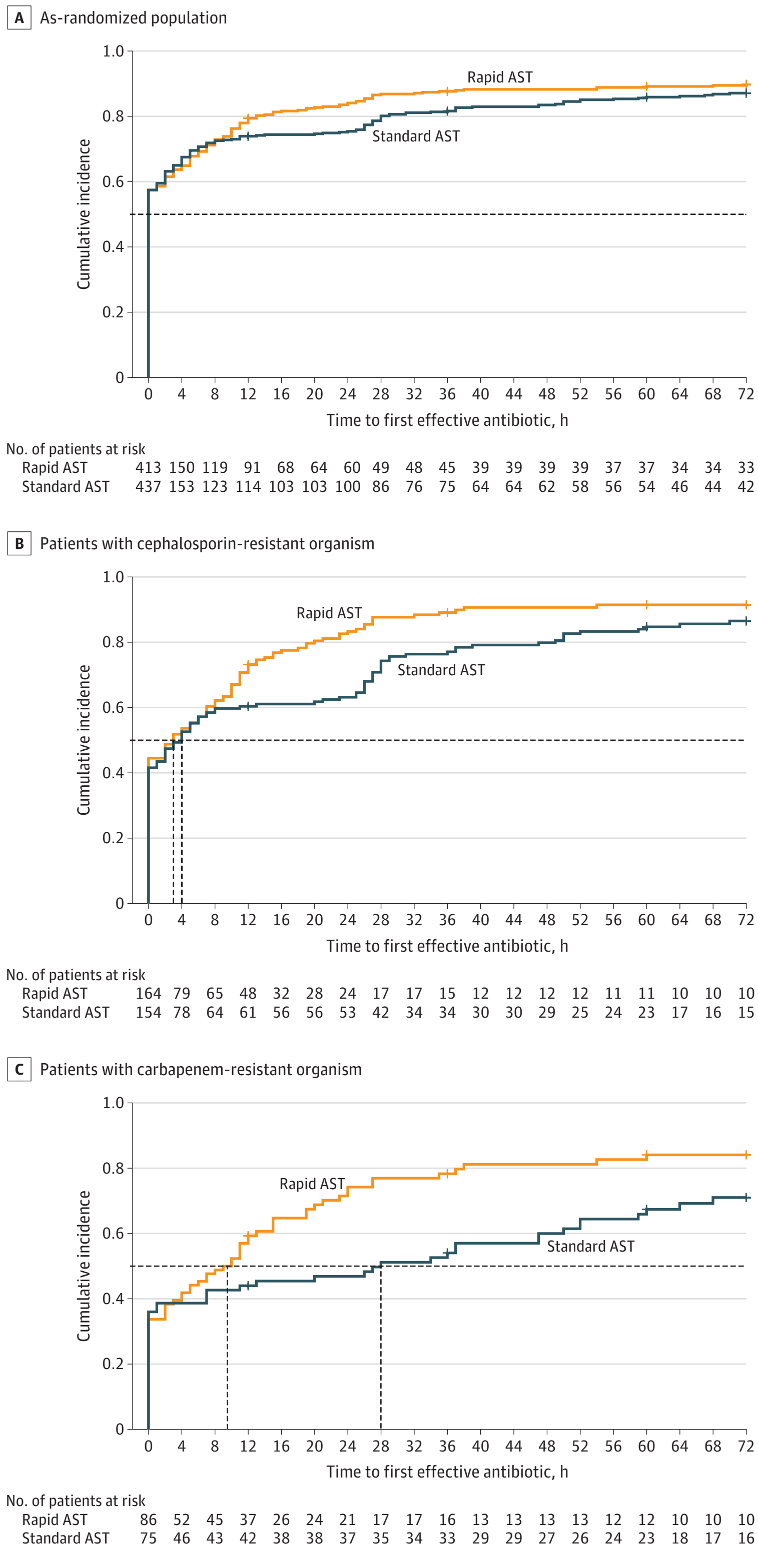
[d] Included relapse of bloodstream infection with the same species causing the index infection, local suppurative complications not present at time of Gram stain result, or seeding distant sites with the same infecting organism after Gram stain result.

[e] Defined as an antibiotic to which the bloodstream isolate was susceptible by standard antibacterial susceptibility testing methods and to which the organism was not intrinsically resistant. Baseline period included time from Gram stain through 2 hours after Gram stain.

[f] Defined in Supplement 3.

obtain aztreonam for several months due to a national shortage of this antibiotic, which is often used for treatment of specific carbapenemase-producing bacteria. Furthermore, first-line therapies used in many countries to treat carbapenem-resistant organisms (eg, sulbactam-durlobactam) are not readily available in India or Greece, from which more than half of participants were enrolled.[43] Approximately 10% to 15% of patients categorized as receiving appropriate therapy in this

Figure 3. Survival Curves of Cumulative Incidence of Time to Receipt of First Effective Antibiotic Within 3 Days After Gram Stain Result

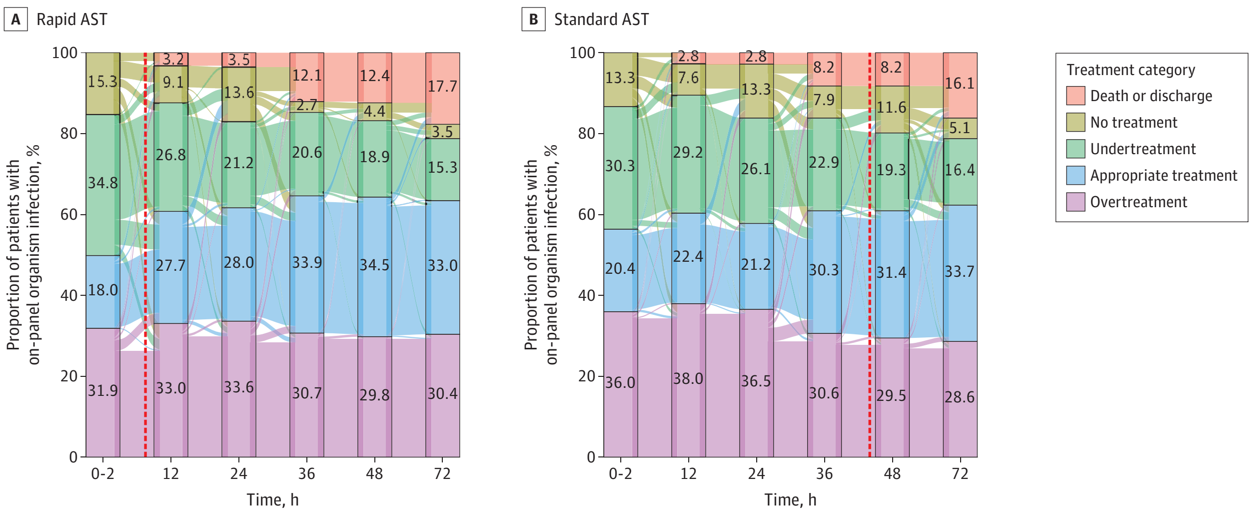


AST indicates antimicrobial susceptibility testing.

Effective antibiotic was defined as receipt of an antibiotic to which the bloodstream isolate tested susceptible using standard AST methods and to which the organism was not intrinsically resistant (eMethods in Supplement 3). Baseline was up to 2 hours after Gram stain result; censored at 72 hours or set to 12 PM on the date of death or discharge. Plus signs indicate censorship. Horizontal dashed lines indicate 50% and vertical dashed lines indicate the time when 50% of participants in a group received the first effective antibiotic. A, Medians were not estimable because more than 50% of patients were undergoing effective therapy at enrollment. B, Median hours in the rapid vs standard AST groups were 3 (95% CI, 0-7) vs 4 (95% CI, 1-7), respectively (difference, −1 [95% CI, −6 to 4]). C, Median hours in the rapid vs standard AST groups were 9.5 (95% CI, 3-13) vs 28 (95% CI, 1-50), respectively (difference, −18 [95% CI, −42 to 6]).

study were receiving colistin- or polymyxin-based regimens, which are older agents not recommended for first-line use by US or European treatment guidelines because they are inferior to newer antibiotics and carry additional toxicities and

Figure 4. Sankey Diagram of Patterns of Antibiotic Treatment in Patients With Monomicrobial, On-Panel Organism Infections



Both the effectiveness and spectrum of antibiotics received were evaluated among participants according to the infecting organism (Supplement 3). Appropriate treatment was defined as receipt of a narrow-spectrum, effective antibiotic, excluding monotherapy with aminoglycosides. Overtreatment was defined as receipt of 1 or more unnecessarily broad-spectrum antibiotics to which the bloodstream isolate was susceptible. Undertreatment was defined as not receiving any effective antibiotics. Among 110 patients categorized as receiving undertreatment at 72 hours, 38 (35%) were receiving piperacillin/tazobactam for ceftriaxone-resistant *Enterobacterales*, 29 (27%) were receiving antibiotics to which their blood isolate was resistant or monotherapy with an aminoglycoside, and the remaining 43 patients (38%) had missing antimicrobial susceptibility results. Among 231 patients categorized as receiving appropriate antibiotic treatment at 72 hours, 30 (13%) were receiving colistin- or polymyxin-based regimens. Treatment categories are further defined in Supplement 3. Dashed vertical lines indicate the median (IQR) time from Gram stain result to AST result and were 7.5 (1.3) hours for the rapid AST group and 44 (23) hours for the standard AST group.

AST indicates antimicrobial susceptibility testing.

adverse effects.[43,44] Equitable access to optimal antibiotic therapies is needed alongside diagnostic innovation to improve patient outcomes. Additionally, patients with antibiotic-susceptible isolates, who comprised more than half of trial participants, were most often receiving effective therapy at enrollment; for these patients, rapid AST facilitated safe antibiotic deescalation and improved judicious antibiotic use, but would have been unlikely to affect mortality.[45] Lastly, initiating optimal therapy a few hours earlier might not affect outcomes of patients with milder forms of sepsis (ie, not septic shock) or with suppurative complications already present at enrollment.[46]

Strengths and Limitations

Study strengths include the pragmatic, real-world implementation of the intervention that allowed sites to incorporate local adaptations for coordination of their laboratory, antimicrobial stewardship, and clinical teams. Additional strengths include the large sample size and enrollment in areas of the world with a high burden of antimicrobial-resistant infections that have not been well represented in prior blood culture diagnostic outcome studies. The rapid AST method used is a phenotypic, rather than genotypic, method for detecting antimicrobial resistance, which enables detection of a multitude of known and novel resistance mechanisms.

This study has limitations. First, sites were heterogeneous in prevalence of antimicrobial resistance and baseline clinical and microbiology practices and implemented the multifaceted intervention differently, which may have obscured benefits in the intervention group. An attempt was made to address this by distributing standardized stewardship and treatment guidance to sites and stratifying randomization by site to balance site-specific factors. Second, study findings may not be generalizable to all institutions because the effect of blood culture diagnostics differs based on a variety of institution-specific factors, including patient complexity, turnaround time of standard culture and AST methods, availability and engagement of antimicrobial stewardship programs, local pathogen prevalence, and resistance rates. For example, *Salmonella enterica* serotype Typhi was observed only in subjects enrolled from India, and this organism was not a target on the rapid AST platform used in this trial. The rapid test may thus have less benefit in regions with a high prevalence of off-target organisms. Third, nearly 20% of enrolled patients had off-panel organisms for which the rapid AST did not provide a result; this percentage was higher than anticipated and may have reduced impact of the diagnostic, although results were similar when the analysis was restricted to the subgroup with on-panel organisms. Fourth, the trial used a bundled intervention of rapid AST together with antimicrobial stewardship review and did not evaluate the independent effects of rapid AST vs antimicrobial stewardship. The trial was designed as such because previous studies demonstrated more optimal antibiotic use when rapid blood culture diagnostics are paired with stewardship rather than used alone.[16,47-49] Fifth, the rapid test did not provide organism identification, which may not be applicable for all laboratory workflows because they vary across institutions.[50] Participating sites likely had more laboratory capacity than many facilities in low-resource settings because they were able to implement a new AST method and use MALDI-ToF mass spectrometry for organism identification. Sixth, antibiotic treatment regimens were not standardized and were determined by local clinicians. Antibiotic availability and cost influenced treatment decisions at some sites. Seventh, the cost-effectiveness of combining rapid AST and stewardship was not evaluated, which is an important area for future research.

Conclusions

Among patients with GNB BSI in diverse health care settings, rapid blood culture AST was not superior to standard testing by DOOR. When considered with other efficacy and safety outcomes, these findings may help inform the use of rapid susceptibility testing in clinical practice.

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